7

An Omnidirectional Antenna with Low RCS

7.1 INTRODUCTION

In Chapter 2, 5, and 6 we considered arrays backed by a groundplane. Such configurations typically produced pencil beams. There is, however, also considerable interest in antennas with omnidirectional radiation patterns. The principles for these will be presented in this chapter. It should be emphasized, however, that we shall limit ourselves to designs with inherently low RCS. That rules out the common simple dipole and monopole—in fact, most omnidirectional antennas.

Furthermore, in Chapter 8 we will show that in order to design a parabolic antenna with a low RCS, it is crucial to use a feed with low RCS. Such a feed is readily obtained by a relatively simple modification of the omnidirectional antenna presented in this chapter. Note that no power is lost in the omnidirectional antenna or in the low RCS feed when transmitting. See also Section 7.6.

We should point out that the RCS level of a parabolic antenna never can attain the low levels possible with an array. They are, however, popular in some camps and will therefore be discussed in Chapter 8.

7.2 THE CONCEPT

In Chapter 2, Section 2.6 we investigated planar arrays of dipoles without a groundplane as shown in Fig. 7.1a. Such configurations have a transmitting

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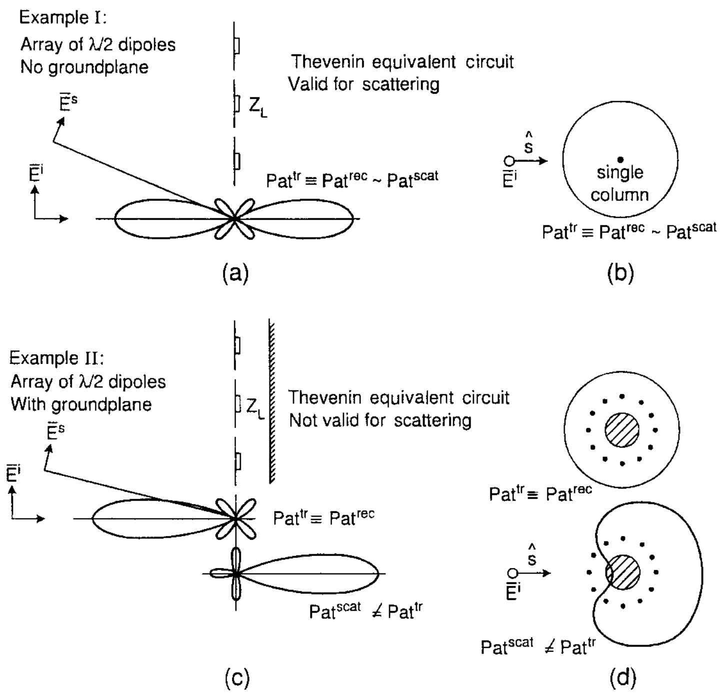

Fig. 7.1 The transmitting, receiving, and scattering pattern Pat^{tr}, Pat^{rec}, and Pat^{scat}, respectively, for various antenna configurations: (a) Array of dipoles without a groundplane, (b) single column of collinear dipoles, top view, (c) array of dipoles with a groundplane, (d) columns of vertical dipoles arranged in a circle around a conducting rod or pipe, top view.

pattern Pat^{tr} comprised of two pencil beams in directions opposite each other (we assume all elements are fed in phase). If we instead consider a single column of dipoles as shown in Fig. 7.1b, the transmitting pattern Pat^{tr} will be omnidirectional. When either one of those two configurations are exposed to an incident plane wave, the RCS will for conjugate match be only $\sim$6 dB below the maximum value obtained when the antenna terminals are essentially short-circuited. This is simply not low enough to be of any practical interest.

Let us instead consider an array of dipoles backed by a groundplane as shown in Fig. 7.1c. From Chapter 2, Section 2.6, we know that the backscattered signal can in principle be "infinitely" low. (For edge effect and fine-tuning see Chapter 5.)

The omnidirectional version of the planar array with a groundplane is shown in Fig. 7.1d. It consists of a conducting rod or pipe surrounded by dipoles arranged in columns. It is well known that if these elements are fed in phase, we will obtain an omnidirectional transmitting pattern Pat^{tr} that essentially can be described as a circle with ripples. The depth and number of these ripples (not shown in Fig. 7.1d) depend on the density of the columns. The closer they are to each other, the smaller the ripples. In fact, when the columns in the limit form a continuous current sheet around the pipe, the transmitting pattern Pat^{tr} simply becomes a pure circle with no ripples at all.

Any antenna exposed to an incident plane wave will receive a signal at its load impedance Z_L. The receiving pattern Pat^{rec} will, according to the reciprocity theorem, always be identical to the transmitting pattern Pat^{tr} as discussed in depth in Chapter 2, Section 2.9. We also there define the scattering pattern Pat^{scat} as the bistatic scattered field reradiated from the antenna when exposed to an incident plane wave. It is obtained by calculating the radiation pattern based on the antenna current under receiving (i.e., scattering) conditions. Many readers will expect this pattern to merely be equal to the receiving pattern (and thus identical to the transmitting pattern). While this may be approximately true for the two examples shown in Figs. 7.1a and 7.1b, it is far from true in general. As an example, consider Fig. 7.1c, as discussed in detail in Chapter 2, Section 2.9. The scattering pattern had its mainbeam pointing in the forward direction and not in the backward direction as does $Pat^{tr} \equiv Pat^{rec}$; see the radiation pattern in Fig. 7.1c. Furthermore, we remind the reader that the transmitting and receiving pattern always are completely independent of the load impedance Z_L (or the generator impedance Z_G). In contrast, the scattering pattern is extremely dependent upon Z_L in the back sector but hardly in the forward direction as illustrated by direct calculations shown in Chapter 5, Figs. 5.2 and 5.3. We finally emphasize that the forward mainbeam of the scattering pattern is not merely equal to the mainbeam of the transmitting pattern rotated 180°. They may in some cases be similar, but in general they are not that simply related. Again, consult Figs. 5.2 and 5.3.

A typical scattering pattern of the omnidirectional antenna is shown in the lower part of Fig. 7.1d. We observe a "main beam" in the forward direction and a low backscattered signal in the back sector. Note that the forward scattering changes little with the load impedance Z_L while the backscatter is very sensitive. For further discussion as well as calculated scattering pattern see Sections 7.4 and 7.6.1.

7.3 HOW DO WE FEED THE ELEMENTS?

As mentioned earlier, we will obtain an omnidirectional transmitting (and thus also receiving) radiation pattern if all the elements are fed in phase. Basically this can be done by using an ordinary harness with simple T connectors as shown for example in Chapter 2, Section 2.12. However, as also shown there, we must prevent coupling between the elements through the feed network since this could lead to high backscatter return. Use of hybrids is a proven way to avoid the calamity.

An example of this approach is shown in Fig. 7.2. Note that due to the hybrids each element will always see the same resistive load regardless of the nature of the incident field.

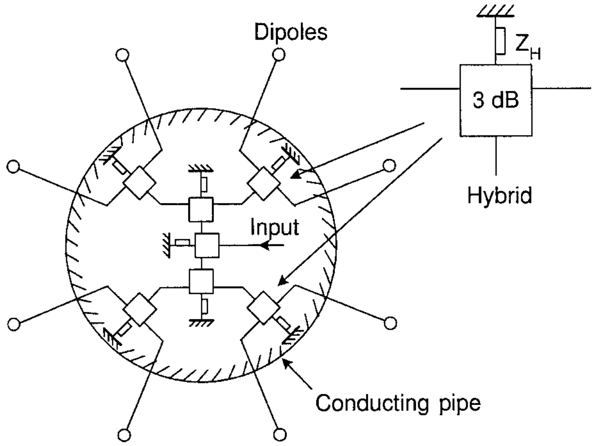

Fig. 7.2 Columns of vertical dipoles arranged in a circle around a conducting pipe, inside which is the harness feeding the dipoles. Use of hybrids ensures that all dipoles are seeing the same load regardless of how the dipoles are excited by the incident field. In other words, the hybrids prevent any coupling between elements through the harness.

Thus, when calculating the scattering patter, we shall model the antenna as a circular array of dipoles loaded with the same load resistances.

7.4 CALCULATED SCATTERING PATTERN FOR OMNIDIRECTIONAL ANTENNA WITH LOW RCS

In Figs. 7.3 and 7.4 we show the calculated bistatic scattering pattern for an omnidirectional infinitely long low RCS array [108]. The diameter of the pipe was around 0.5λ at the center frequency, while dipoles were located on a circle with diameter around 0.75λ. The load resistors were not conjugate-matched but close to it (remember, only planar arrays become invisible for conjugate matching). The scattering pattern at $f = 4$, 6, and 8 GHz are shown in Fig. 7.3, and similarly the patterns at 10, 12, and 14 GHz are shown in Fig. 7.4. We note that the forward-scattered field changes very little with frequency. However, as we would expect, the field scattered in the backward sector varies significantly with frequency and matching.

7.5 MEASURED BACKSCATTER FROM A LOW RCS OMNIDIRECTIONAL ANTENNA

The patterns above were first calculated in a thesis by Jeff Hughes [108], a former student and now very successfully working for the U.S. Air Force. While there never was any doubt in the author's mind about the validity of Jeff's results, there are always those who refuse to believe anything contrary to their

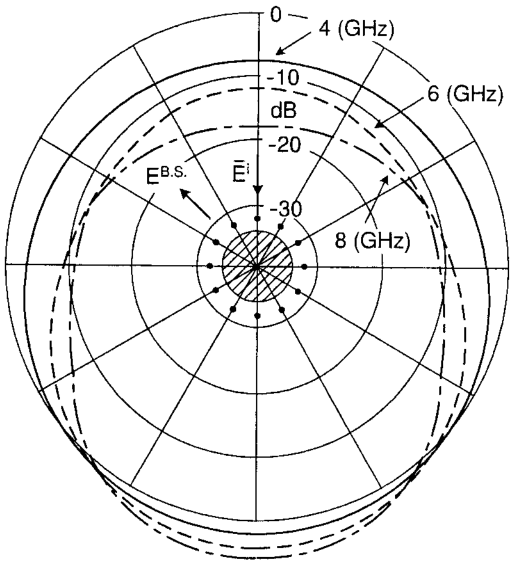

Fig. 7.3 The calculated bistatic scattering pattern Pat^{scat} for a circular array of dipole columns arranged around a conducting rod or pipe. Frequencies 4, 6, and 8 GHz [108].

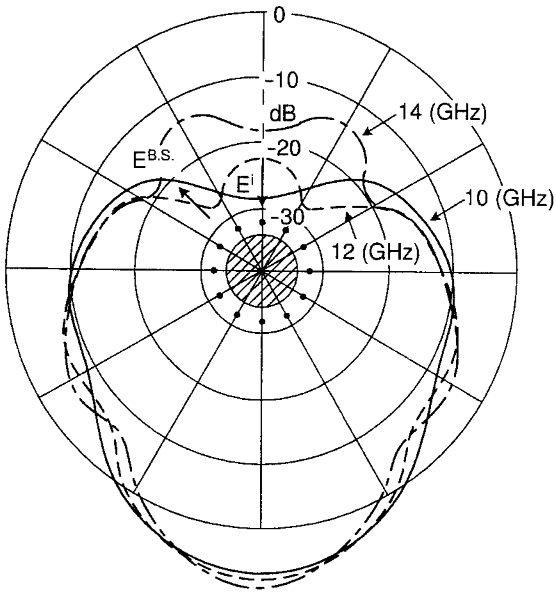

Fig. 7.4 The calculated bistatic scattering pattern Pat^{scat} for a circular array of dipole columns arranged around a conducting rod or pipe. Frequencies 10, 12, and 14 GHz [108].

"intuition" (often misguided by insufficient training). They simply demand to see actual measurements.[1]

Thus, the investigation was continued and greatly extended in the dissertation by Jay Simon [109]. He, assisted by other students, designed, built, and measured an actual antenna. Because the elements at X-band frequencies were rather small, it was for tolerance reasons decided to change the frequency band to the S-band region. Jay very carefully calculated the optimum loads to be placed at the terminals of each dipole (this would simulate the input impedance of receivers, amplifiers, or the generator impedances of the transmitters; see also Chapter 2, Sections 2.10 through 2.12). Again, note that the optimum value of the load impedance was slightly different than the conjugate match. The reason for this is simply that this antenna is nonplanar. It implies that the residual scattering no longer is zero but small.

The measured array consisted of a conducting pipe with diameter 4 cm while the vertical dipoles were photoetched on a thin sheet of substrate rolled up in a cylinder with 6.7-cm diameter. The length of each of the vertical dipoles were 3.6 cm, and there were 20 of them in each of the 10 columns. That corresponds to a total array length of 87 cm.

The measured monostatic RCS of this finite array as a function of frequency is shown in Fig. 7.5. We show three RCS curves where the field is incident in the equatorial plane.

a. One where the entire circular array is covered with a layer of foil fitting snugly around all dipoles.
b. Also, as a further verification, the backscattered signal from the center pipe alone without any dipoles or foil.
c. And finally, the backscattered field when all the loaded dipoles are placed around the center pipe simulating the actual antenna.

We note that the measured RCS level is as much as 40 dB below the foil-covered antenna at one frequency. That is in fact fairly close to what Jay calculated. Furthermore, the relative 13-dB bandwidth is about $5.2/2.8 = 1.9$—that is, approximately an octave wide. Note that this result is obtained for fixed load impedances without any broadband matching as discussed in Appendix B. Further increase in bandwidth can be obtained by using more elements.

We also show in Fig. 7.6 the return from the same antenna but for horizontal polarization with and without foil. As one would expect, the return is similar to the pipe alone for vertical polarization.

Finally, in Fig. 7.7 we show the backscattering from an antenna with horizontal curved elements. The pipe diameter was 4 cm, and the diameter of the circle with the dipoles was 6.7 cm with a total array length of 87 cm. The depth of the null is similar to the vertical case, but it is more narrowbanded. Again, we emphasize

[1] Jeff's approach was completely rigorous in that he expanded everything including the field from the pipe into Hankel functions with origins at the center.

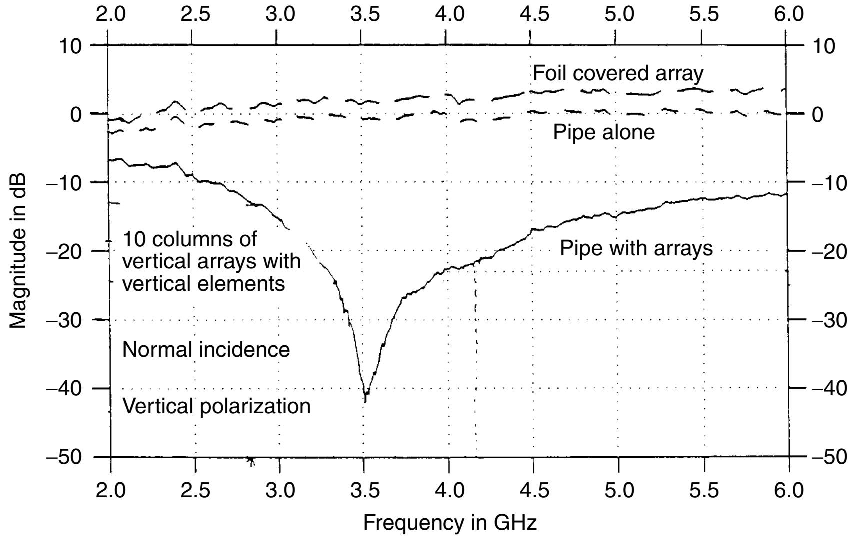

Fig. 7.5 The measured backscattering from an array of resistively loaded vertical dipoles arranged around a conducting pipe as shown in Figs. 7.3 and 7.4. Three curves are shown: one where the entire array is covered with foil (baseline) one of only the pipe without dipoles, and one where the pipe is surrounded by resistively loaded dipoles. Polarization of incident and reflected field: vertical [109].

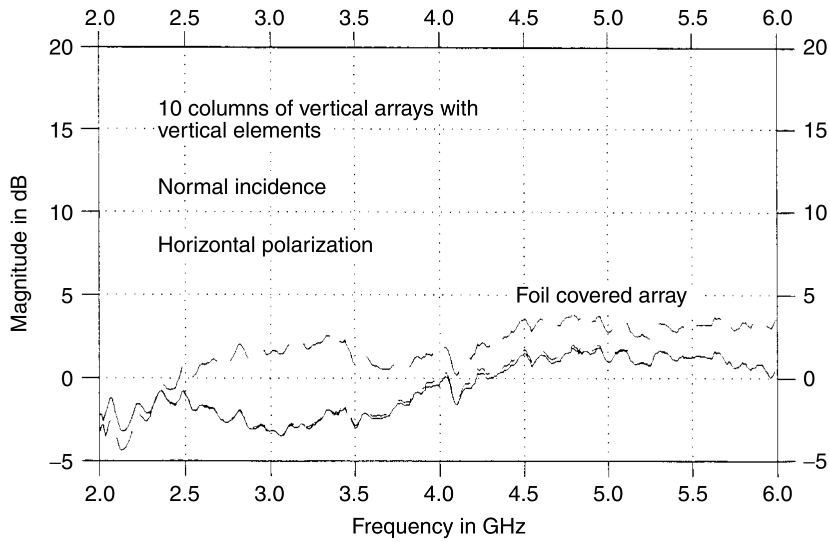

Fig. 7.6 Same case as shown in Fig. 7.5 but where the incident and reflected fields are horizontally polarized [109].

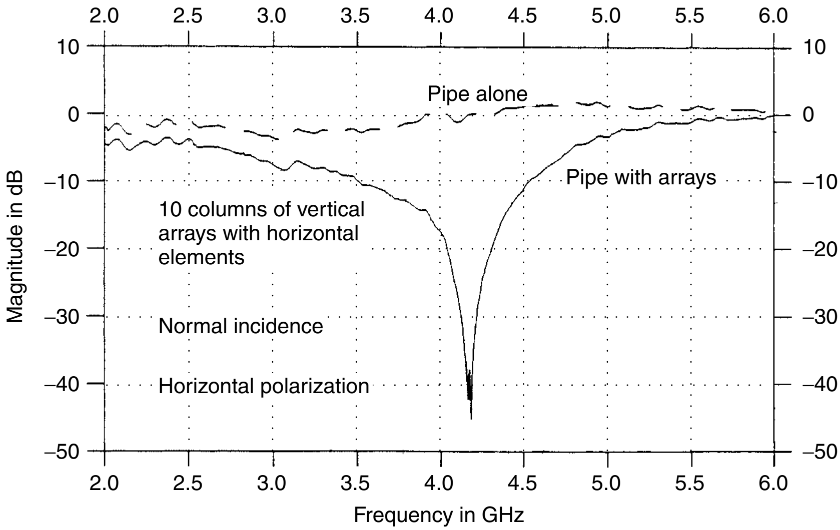

Fig. 7.7 The measured backscattering from a circular array of horizontal resistively loaded dipoles photoetched on a thin substrate curved around a conducting pipe. Two curves are shown: one of the pipe alone without dipoles and another with the resistively loaded dipoles. Polarization of incident and reflected field: horizontal [109].

that these results were obtained for constant load resistors without the use of a broadband matching technique.

Incidentally, both array models can still be viewed in the author's office. However, some ravage from rough handling over the years is observable.

7.6 COMMON MISCONCEPTIONS

7.6.1 On the Differences and Similarities of the Radiation Pattern

The transmitting, receiving, and scattering patterns are often the subject of some misunderstanding. Let us consider the omnidirectional low RCS antenna in particular.

Actually, the transmitting pattern is straightforward to comprehend. All elements are fed with the same in-phase currents via the harness including the hybrids. Surely the radiation pattern will basically be omnidirectional with ripples as discussed earlier. The receiving pattern is more complicated. When a plain wave is incident upon the antenna, there will be stronger currents on the lit side in the front than on the shadow side in the back. This imbalance might cause some power to go into the hybrid loads rather than the receiver load. Furthermore, the elements on the shadow side will deliver less power than those at the front. All of this can lead one to believe that the antenna is less efficient when receiving than when transmitting.

This is not the case. We merely have to apply the reciprocity theorem between the main terminals of our omnidirectional antenna and an external test antenna. When transmitting from the array and receiving at the test antenna, no loss will occur in any of the hybrid loads. Conversely, if we transmit from the test antenna and receive on the array terminals, we will certainly lose some power in the hybrid loads, but the total transmission loss must be unchanged whether we transmit or receive. What happened is simply that the elements on the lit side combine into an antenna with higher directivity but the same gain as the omnidirectional case. Remember, the reciprocity theorem deals only with the *terminal* voltages and currents. It says absolutely nothing about the current *distribution* under receive and transmit conditions. While they may be the same in some cases, they are radically different in this case. When this asymmetric receiving current is integrated over all the elements, we will obtain the asymmetric scattering pattern shown in Figs. 7.3 and 7.4.

Finally, there are those who wonder how the scattered field in the forward direction can be maximum when everyone knows that we have a "shadow" there. Well, as explained already in Chapter 2, Section 2.2, and Chapter 5, Section 5.3, these individuals are simply thinking about the *total* field. Recall that this is the sum of the scattered and the incident fields. They will partly cancel each other in the forward sector and create a shadow as expected. In other words, the incident field will primarily be absorbed by the antenna when matched, and a relatively weak *total* field will be observed in the forward scattering sector. See also Chapter 5, Figs. 5.2 and 5.3.

7.6.2 How You Can Lock Yourself into the Wrong Box

A highly respected professor in electromagnetics once told the author that pursuing the development of a low RCS omnidirectional antenna most likely was a waste of time.

He based his statement on the following argument: Consider a dipole. When a load impedance is connected to its terminals, a current will flow through the load and thus also through the antenna wire (at least in the vicinity of the terminals). And since that current will reradiate, we will observe a significant RCS.

His argument was flawless in every respect except one: He limited himself at the onset to a simple dipole. It might have been the first time the author replied: "If you have a problem you cannot solve, change the problem!"

Today we know of course that a circular array placed around a pipe acting as a groundplane can be designed to have a low RCS without any significant loss. And where did that come from? Well, it came from the planar arrays with a groundplane, as discussed in great length in Chapters 2 and 5. And where did this seemingly hopeless idea come from? From the fundamental antenna concept that a large aperture backed by a groundplane can absorb *all* the incident energy; that is, *nothing* will be left to be reflected in the back direction (concerning edge effect; see Chapter 5). Oh, if we could just get our young engineers to study fundamentals again!

7.7 CONCLUSIONS AND RECOMMENDATIONS

Many readers intuitively assume that it is inherently impossible to design an omnidirectional antenna with low backscatter.[2] This assumption is based upon considering only a simple dipole as discussed in Common Misconceptions, Section 7.6.2. However, we have shown in this chapter that just as a planar array of dipoles backed by a groundplane has a very low backscatter, see Chapters 2 and 5, so can a circular array of dipoles placed around a conducting pipe. It was remarkable that the measured RCS at one frequency was $\sim$40 dB below the same array covered by foil. A 13-dB reduction was obtained over a bandwidth of almost an octave. It should also be noted that the load resistors were somewhat different than conjugate match. This is simply because the curved surface has a small residual scattering component in contrast to the planar surface which has none (in principle). No broadband technique as illustrated in Chapter 6 and Appendices A and B was applied.

It is recommended to further investigate the effect of the "groundplane" in the form of the central pipe just as we did in the planar case discussed in Chapter 6. No doubt, with our increased knowledge about design today, we could surely obtain a better and more broadbanded cancellation of the element and the groundplane reactances.

Finally, just like a dielectric slab placed in front of a planar dipole array can increase the bandwidth (see Chapter 6, Sections 6.5 through 6.11), we strongly expect the same to be true when a dielectric sleeve of proper thickness and dielectric constant is placed around the dipole elements on the outside. In fact, Simon [109] already incorporated the effect of dielectric on the outside, but no attempt was done to design for an increase of the bandwidth. After all, dissertations should be unclassified (although the author's dissertation was classified for many years).

[2] All antennas receiving real energy from an incident wave will have a strong scattered field in the forward direction. However, the total field being the sum of the incident and the scattered field is weak.